Cargill-SMU Ocean Transportation Datathon 2026:
A Data-Driven Strategy for Voyage Employment

The Data Minions

Introduction

This report presents a data-driven voyage employment strategy for four Cargill Capesize vessels and three committed Cargill cargoes, with the objective of maximizing portfolio profit and Time Charter Equivalent (TCE) under prevailing market conditions.

We developed a Freight Calculator that incorporates voyage revenues, operating costs, bunker consumption, port costs, and delays. To better capture real-world uncertainty, we enhanced the deterministic freight model with a machine learning-based risk adjustment to account for shipping risks such as port congestion and adverse weather.

Our base-case recommendation assigns:

1. Each Cargill vessel to its most profitable next voyage
2. Each committed cargo to the vessel (Cargill-owned or market vessel) that maximizes risk-adjusted profit.

Vessel	Type	Cargo	Route	TCE	Profit	Days	Completion	Rationale
ANN BELL	Cargill	COMM_1	Kamsar→Qingdao	$28,280	$1,541,323	93.2	May 29	Optimal positioning, highest TCE, makes laycan
OCEAN HORIZON	Market	COMM_2	Hedland→Lianyungang	$22,084	$194,699	30.7	March 31	Geographically efficient, makes laycan, fastest turnaround
CORAL EMPEROR	Market	COMM_3	Itaguai→Qingdao	$37,076	$1,128,258	77.1	May 21	Highest TCE overall, makes laycan

Table 1: Recommended Vessel - Cargo Allocations and Voyage Economics

We further conducted scenario analyses on bunker prices and port delays to identify threshold conditions under which alternative voyage options become optimal. Finally, we implemented a Generative AI-powered chatbot to enable interactive querying of voyage recommendations and scenario outcomes.

Recommended cargo-vessel allocation and rationale

Combination 1: Ann Bell → Committed_1
This combination is the most economically attractive option among all evaluated scenarios. **ANN BELL** is already positioned at Qingdao, which is the discharge port for Committed_1, eliminating ballast risk at the start of the voyage. The cargo of 180,000 MT of bauxite fits comfortably within the vessel's 180,803 DWT capacity, ensuring full utilization without operational constraints.

The voyage generates an estimated USD 1.54 million in profit, corresponding to a 3.7% net margin and the highest TCE of USD 28,280/day across all combinations. Economics are further strengthened by the contractual structure, under which all port costs are borne by the charterer, and a lower commission rate of 1.25%, compared to 3.75% for other cargoes. From an operational standpoint, the vessel has a 94.2% probability of making the laycan, arriving on 4 April, two days after the window opens. The vessel also carries a 2.5-day buffer, reducing the risk of missing laycan. The primary risk is the long ballast leg to West Africa (11,615 nm) and potential weather disruption in the Indian Ocean during March–April. However, this is partially mitigated by the vessel's healthy bunker position (401 MT VLSFO ROB), which reduces exposure to bunker price volatility. Overall, this combination offers superior risk-adjusted returns and is strongly recommended

Combination 2: Ocean Horizon → Committed_2
This combination is selected primarily for operational efficiency and positioning, rather than absolute profitability. **OCEAN HORIZON's** current position at Map Ta Phut, Thailand provides a short and efficient ballast leg to Port Hedland (3,040 nm via Singapore). The vessel departs on 1 March, arrives on 10 March, and comfortably meets the laycan window of 7–11 March.

The voyage generates a modest USD 195,000 profit, with a 1.4% net margin. While margins are thin, the voyage covers all operating costs and contributes positively to overhead. Total voyage duration is only 30.7 days, allowing the vessel to be redeployed quickly for Q2 opportunities, which is strategically valuable. Cargill's strong operational familiarity with Australian iron ore ports further reduces execution risk. Although port costs are high (USD 380,000) and materially erode margins, this is partially mitigated by fast loading rates of 80,000 MT/day, which minimize port stay and associated costs. The vessel's 266 MT VLSFO ROB also limits immediate bunker exposure. In summary, this is a low-risk, low-return employment that prioritizes schedule efficiency and fleet positioning.

Combination 3: Coral Emperor → Committed_3
This combination involves chartering a market vessel to fulfill Committed_3 and delivers strong risk-adjusted returns. **CORAL EMPEROR** arrives on 23 March, comfortably making the laycan with nine days of buffer, allowing for potential waiting time without laycan risk. The voyage completes on 21 May 2026, aligning well with the Q2 trading horizon. The voyage generates an estimated USD 1.13 million in profit, with a TCE of USD 37,076/day. After accounting for the hire rate of USD 22,436/day (Q2-26 5TC), the net TCE remains attractive at USD 14,640/day, making this a commercially sound chartering decision.

Although this option requires exposure to the spot charter market, the comfortable laycan buffer and solid economics justify the use of a market vessel rather than deploying a Cargill-owned ship. Overall, this combination provides a balanced trade-off between profitability and operational certainty, making it the preferred option for Committed_3.

In conclusion, the three vessel–cargo combinations were selected using a conservative, risk-managed strategy that prioritised 100% laycan compliance, customer relationship protection, and fleet flexibility over absolute profit maximisation. By ensuring zero laycan risk and limiting exposure to market vessels, the allocation delivers high execution certainty while preserving two Cargill-owned vessels for potential Q2 opportunities. This approach accepts lower profitability on COMM_2 and higher charter costs for **CORAL EMPEROR**, resulting in total profit of USD 2.86M, approximately USD 580K (17%) below the theoretical optimum, but USD 450K (19%) above a conservative baseline. Overall, the strategy reflects sound commercial judgement, trading marginal upside for reliability, reputational protection, and operational simplicity, which are critical in managing long-term customer relationships in committed cargo operations.

Results of scenario analyses and threshold insights

Port delay in China

With increasing discharge port delays on **ANN BELL**, its profitability steadily declines. At an additional delay of approximately 11.8 days, ANN BELL's profit falls to about $1,082,548, dropping below **OCEAN HORIZON's** profit of $1,094,274. Beyond this point, **ANN BELL** is no longer the optimal vessel choice. **ANN BELL's** profit decreases from $1,232,786 under the base case to $1,080,002 at 12 days of delay, $1,054,538 at 14 days, $1,051,737 at 14.22 days, and further to $978,146 at 20 days, continuously staying as the less optimal choice.

ANN BELL's advantage is time-sensitive because its lower hire rate only holds as long as delays are minimal. Each day of discharge delay costs approximately $12,732, made up of $11,750 in daily hire and about $982 in idle fuel consumption. Over 11.8 days, this translates to an additional cost of roughly $150,237. As a result, **ANN BELL's** profit erodes from an initial $1,232,786 to about $1,082,549. While **ANN BELL** initially benefits from a lower hire rate compared to **OCEAN HORIZON** ($11,750 versus $15,750 per day), any prolonged delay quickly wipes out this advantage, explaining why its profitability falls below **OCEAN HORIZON** once delays reach around 11.8 days.

Discharge delay	Profit	TCE ($/day)	vs OCEAN HORIZON
0 days	$1,232,786	$26,077	$138,512
5 days	$1,169,126	$24,591	$74,852
10 days	$1,105,466	$23,260	$11,192
11.8 days	$1,082,548	$22,814	-$11,726
12 days	$1,080,002	$22,765	-$14,272
15 days	$1,041,806	$22,060	-$52,468

Table 2: Impact of Discharge Port Delays on ANN BELL profitability vs OCEAN HORIZON

Therefore, the breakeven threshold occurs at roughly 11.8 days of additional discharge port delay, after which **OCEAN HORIZON** becomes the more profitable option.

<u>Bunker Price Increase at all ports</u>
We tested the sensitivity to vessel selection to increases in VLSFO prices, holding all other parameters constant. Across all tested fuel price scenarios (up to a 54% increase in VLSFO prices), **ANN BELL** remains the more profitable option.

VLSFO Price (USD/MT)	ANN BELL Profit (USD)	OCEAN HORIZON Profit (USD)	Profit Gap (USD)
491 (Base case)	1,232,786	1,094,274	138,512
662 (+35%)	611,700	503,310	108,390
758 (+54%)	263,020	173,469	89,551

Table 3: Comparison between ANN BELL and OCEAN HORIZON of how VSPFO affects profit

The main reason for why the recommendation does not flip is a lower daily hire rate. While **ANN BELL** consumes only around 196 MT more VLSFO than **OCEAN HORIZON,** it benefits from a USD 4,000/day lower hire rate. Over the full voyage duration, this translates to around USD 344,000 in hire cost savings, which offsets the additional fuel expense even under elevated bunker prices. Hence, there is no reasonable threshold that changes the vessel recommendation.

Key Assumptions

All voyage economics are based on Q2-26 market, operational, and financial assumptions. The market charter cost for **CORAL EMPEROR** is derived from Baltic Exchange Q2-26 5TC FFA rates of USD 22,436/day, assuming full vessel availability, no off-hire events, and acceptance of early arrival and waiting time; deviations in actual market rates present both upside and downside risk. Forward bunker prices are assumed to be accurate at the time of bunkering, with VLSFO prices between USD 467–491/MT, no major supply disruptions, and bunkering occurring at planned ports without delay. Voyage routing and distances rely on system pathfinding using standard maritime routes, assuming no canal restrictions and excluding weather routing impacts. Operationally, all vessels are assumed to sail at economical speed, prioritizing fuel efficiency, with no weather delays, port congestion, or mechanical failures in the base case. Standard port turn times and Per Weather Worker Day, Sundays and Holidays Included (PWWD SHINC) working terms are applied, with terminals operating continuously. Vessel fuel consumption is assumed to align with declared performance, supported by adequate ROB levels for planned bunkering strategies. Financial results for Cargill-owned vessels reflect internal hire rates for transfer pricing rather than market levels, with performance assessed using Net TCE. Commission deductions follow contractual terms. Strategically, the allocation prioritizes laycan reliability, customer relationships, and fleet flexibility over marginal profit maximization, retaining unused vessels for potential Q2 market opportunities.

Conclusion

This report presents a risk-managed, data-driven voyage employment strategy developed for the datathon, focusing on the optimal deployment of four Cargill Capesize vessels and three committed cargoes under Q2-26 market conditions. The objective was to maximise portfolio profit and TCE while ensuring high execution certainty.

A comprehensive freight calculator was built to evaluate voyage revenues, operating costs, bunker consumption, port charges, and delays. To better reflect real-world uncertainty, the model was enhanced with a machine learning based risk adjustment, accounting for factors such as port congestion and adverse weather. Scenario analyses on bunker prices and port delays were also conducted to identify conditions under which alternative voyage options become optimal. In addition, an AI powered chatbot was implemented to enable interactive exploration of voyage recommendations and scenario outcomes.

Based on risk-adjusted economics and operational feasibility, three vessel-cargo combinations were recommended. **ANN BELL** was assigned to Committed_1, delivering the strongest performance with the highest TCE and minimal operational risk due to favorable positioning and contractual terms. **OCEAN HORIZON** was allocated to Committed_2 as a low-risk, low-return option that prioritizes schedule efficiency and strategic fleet positioning. Committed_3 was fulfilled using a market vessel, **CORAL EMPEROR**, which offered attractive risk-adjusted returns and ample laycan buffer despite exposure to spot charter rates.

Overall, the strategy prioritizes 100% laycan compliance, customer relationship protection, and fleet flexibility over absolute profit maximization. The resulting total profit of USD 2.86 million is approximately 17% below the theoretical maximum, but 19% above a conservative baseline, reflecting a deliberate trade-off between marginal upside and operational reliability. This approach demonstrates sound commercial judgement and aligns with the long-term priorities of committed cargo operations.